



SENIOR THESIS IN MATHEMATICS

A Mathematical Supplement For First-Year Physicists

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Chapter 1

Introduction

This thesis presents the development, implementation, and evaluation of a set of mathematical supplement modules designed for first-year physics students at Pomona College. These modules aim to bridge the gap between students' mathematical preparation and the quantitative demands of the physics curriculum, addressing a key component of academic support identified by the American Institute of Physics TEAM-UP task force. By providing targeted mathematical content that complements the first-year physics sequence, this work seeks to improve student understanding of fundamental concepts while reducing barriers to success in the physics major.

The document is organized into five chapters. Chapter 2 examines the history of the physics curriculum at Pomona College, tracing the evolution of the introductory sequence and explaining how recent changes like the introduction of Physics 139 (Mathematical Methods of Physics) represent efforts to provide better academic support for physics majors. Chapter 3 outlines the methodology behind the mathematical supplement modules, discussing the assumed mathematical background of first-year students and detailing the process of creating multimodal learning resources that incorporate both written explanations and visual animations. Chapter 4 presents a detailed analysis of the three modules implemented during the quantum mechanics unit of Physics 70, examining student responses and evaluating the effectiveness of different module components through word frequency analysis. Chapter 5 proposes recommendations for future implementation of these supplements across the physics curriculum and suggests enhancements to data collection methods that would allow for more

robust evaluation of their impact on student learning and retention.

Through this work, I hope to contribute to the ongoing efforts of the Pomona College Physics Department to create an inclusive and supportive learning environment that enables students from all backgrounds to thrive in the physics major.

Chapter 2

History of Problem

To contextualize the main goals of my thesis, we must first discuss the curricular trajectory of the first-year physics major. Prior to the fall of 2009, all introductory physics courses were served as a two-semester course: they were called Physics 51a and 51b. Anyone who needed to take physics, whether it was pre-med students or potential physics majors, were grouped into the classes. Hence, there were a wide range of student maturities, with pre-meds primarily in their junior year of college and potential physics majors in their first year. According to the figures shown by Professor Thomas A. Moore at the Winter 2015 American Association of Physics Teachers (AAPT) meeting, there was an average of 65 enrollments for Physics 51a and about 40 for Physics 51b [\[1\]](#).

It should be emphasized that the structure of the 51a/b sequence was not especially traditional either: instead of covering mechanics in the first semester and electromagnetism in the second, they followed Professor Moore's *Six Ideas* textbook, covering conservation, Newtonian dynamics, and special relativity in the first semester. In the second semester, they covered electromagnetism, quantum physics, and thermal physics.

Unfortunately, having a standardized curriculum for both junior pre-med students and first-year physics majors led to dissatisfaction all around. Despite having the course material skewed towards pre-med students, they did not perceive the course to be well-designed for them. Simultaneously, first-years did not feel like their intellectual needs were being addressed. Consequently, many migrated to other majors before they had a chance to explore

more of what the major had to offer. Additionally, many pre-meds took physics because they needed it for the MCAT, a qualifying exam for medical school. Hence, they were less passionate about the material than first-year physics majors, which significantly affected the course atmosphere. The first-year students also noticed how hardworking and capable the junior pre-meds were, leading them to feel left behind at times and doubt their own capabilities. Evidently, the learning outcomes were not as intended: something had to be altered about the course structure.

After considerable discussion amongst the physics department, Physics 51a/b was split into a pair of two-semester courses: Physics 41 and 42 for pre-med students, and Physics 70/71/72 for potential physics majors (Physics 71/72 are each half-courses). The course material was altered slightly as well and shuffled. Physics 41/42 would use Knight's textbook on physics, covering more traditional introductory physics, while the 70 sequence would continue using the *Six Ideas* textbook. In the fall, Physics 70 covers relativity, quantum physics, and thermal physics. In the spring, 71 would cover conservation laws and Newtonian mechanics, and 72 would finish the introductory sequence with electromagnetism.

Such restructuring allowed students to explore topics in physics that very few high school students would have been exposed to, and therefore gave everyone a chance to feel challenged by new material. The three units presented in Physics 70 also didn't emphasize a mastery in college-level math, the only co-requisite being Calculus I, making material accessible to those that didn't have a chance to take higher-level math in high school.

This change was greatly welcome. The year 51a/b was abandoned for the pair of two-semester courses, enrollment in physics classes practically doubled. It continued climbing year-on-year, and by 2014, there were nearly 150 students enrolled in Physics 41 and Physics 70, with about three-quarters taking additional classes in the second semester. Additionally, from 2010 to 2014, the average number of graduating physics majors increased to 11 as opposed to 6-7 between 2006 and 2010. At present, the physics department graduates almost 20 students yearly.

In addition to the split of 51a/b into the 70 and 41/42 sequence, the physics department has more recently introduced PHYS139, a course titled "Mathematical Methods of Physics".

The inspiration was a document published by the American Institute of Physics (AIP) in 2019 titled “The Time is Now: Systemic Changes to Increase African Americans with Bachelor’s Degrees in Physics and Astronomy”, where the AIP National Task Force to Elevate African American Representation in Undergraduate Physics and Astronomy (TEAM-UP) examined the reasons for the persistent underrepresentation of African Americans in physics and astronomy in the US as measured by bachelor’s degrees in such fields [2].

After conducting a major research study that included student surveys, department chair surveys, interviews with African American students amongst other methods, the TEAM-UP task force identified five factors responsible for the success or failure of African American students in physics and astronomy: Belonging, Physics Identity, Academic Support, Personal Support, and Leadership and Structures.

In pioneering the Mathematical Methods in Physics course, the physics department at Pomona decided to act on the Academic Support pillar. From 2019, Professors Newshaw Bahreyni, Elijah Quetin, Philip Choi, Janice Hudgings, Edray Goins (of the math department), and a handful of students began discussing how to best academically support students pursuing the physics major, in particular reducing the barriers to graduate. In the following years, Professors Bahreyni, Quetin, and Goins began designing the Math Methods course. The main consequence of this course was for those who take it to avoid taking numerous math courses, reducing the number of requirements to graduate while maintaining a similar level of mastery of mathematical concepts used in physics. Additionally, this course would let physicists avoid taking courses like Linear Algebra, which emphasized purely mathematical proofs over calculations, the latter mattering more to physicists [3].

In the spring of 2023, Mathematical Methods in Physics was offered for the first time and had an enrollment in the teens. At present, the efficacy of this class is still being evaluated. As an intermediate level mathematical methods course, though, it reviews numerous concepts rapidly, such as Calculus I/II and matrix mechanics. As such, many physicists may benefit from getting a first pass at such material before they take the course, ideally in their first year, when work is not as busy as in subsequent years of college.

Chapter 3

Methodology

3.1 Preliminaries

In this thesis, I want to address the following question: What can be done for aspiring first-year physicists so they can be more mathematically prepared for the physics major?

To start answering this question, I looked back at the first-year physics curriculum, including Physics 70, 71, and 72. According to their syllabi, the courses are designed to be calculus-based [4, 5]. Hence, it is safe to assume that at least some calculus is known, up to the level of Calculus I. Alternatively, if they do not already have this knowledge, they should be co-registered for Calculus I or above. This requirement also assumes the student is familiar with precalculus, but the notion of which concepts that embody are slightly more tenuous. For a working definition, I will assume precalculus covers the following topics:

- Basic knowledge of the elementary functions.
 - This includes monic functions, polynomials, rational functions, power functions, trigonometric functions such as sine, cosine, tangent, their inverses, and reciprocals, exponential functions, logarithmic functions, the absolute value function, and the sigmoid function.
 - Specific properties include domain and range and what they look like graphically. More advanced properties like asymptotes/limit behavior and specific parameters of periodic functions, like period or wavelength, are not necessarily assumed.

- By extension, logarithms and exponentials. Logarithm and exponential rules, such as multiplication as addition and change of base, are somewhat assumed. Students should at least have seen these rules before and only require a brief refresher.
- Conic sections and trigonometry.
 - Students should at least have seen the trigonometric functions, including sine, cosine, tangent, their inverses, and reciprocals.
 - They should also be familiar with the unit circle and the basic trigonometric identities, although these concepts will be reviewed where necessary.
- Knowledge of what vectors are.
 - Students are expected to have heard of dot products and have a vague idea of what they are (this does not apply to cross products), and students should be familiar with how to add and subtract vectors. Additionally, students are assumed to have heard of other linear algebraic concepts like matrices and determinants, but not necessarily know how to manipulate them.

For further motivation, the reason why I am interested in incorporating a mathematical supplement in the first year is because students will encounter increasingly dense math as they tackle the core courses in physics – classical mechanics, electromagnetism, quantum mechanics, and statistical mechanics. To help visualize the pipeline of courses students typically take, I created a concept map in LucidChart that lists both the physical and mathematical concepts students encounter when taking each course, as well as arrows to denote the sequential advancement of courses. The physical and mathematical concepts for each class were obtained by reading various syllabi sent to me by the most recent instructors for each class and their associated textbooks [\[6, 7, 8, 9, 4, 5, 10, 11, 12, 13, 14, 15\]](#).

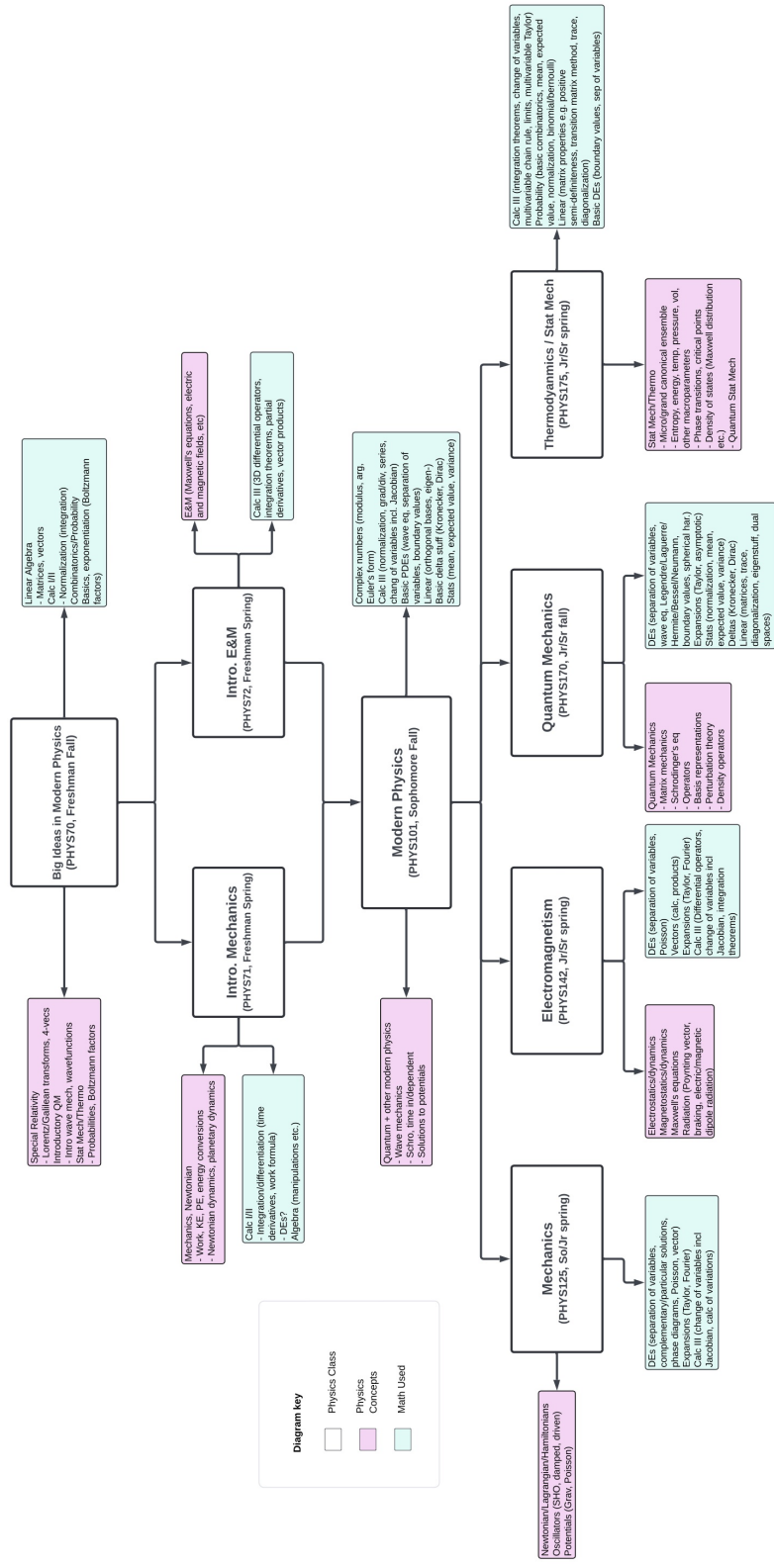


Figure 3.1: A concept map highlighting the typical physics sequence and core courses a physics major at Pomona can expect. Physics concepts are highlighted in pink, while relevant math concepts are highlighted in blue.

With this assumption, I am interested in determining whether the contributions I make are effective in helping students mathematically prepare for the physics major. To address this aim, my aims are as follows:

1. First, I want to make sure all the mathematics that *is* covered in the first-year sequence is absorbed by the students. To this end, I want to ensure that I review any mathematical concepts that are potentially new to the students. For example, in Physics 70, the dot product is introduced when discussing inner products of quantum state vectors. The dot product is fundamentally important to a physicist. Later in the first-year curriculum, it appears many times; in Introductory Mechanics (Physics 71), it appears in integrals when calculating work, and in Introductory Electricity and Magnetism (Physics 72), it appears when calculating electric fields.
2. Additionally, I want to expose students to mathematics that may not be covered in the first-year curriculum but is nonetheless relevant to the physics concept they are learning and will appear in future physics courses. For example, Taylor series are not formally covered in Physics 70 and is a topic generally found in Calculus II, but it is a useful tool in virtually all upper-division physics courses. Therefore, in my work, I introduce Taylor series and motivate their utility through exercises and connections to things students have seen in class, such as the binomial approximation being a linear truncation of the Taylor series of $(1 + x)^n$, as well as when it is appropriate to use a Taylor series in physics problems.

3.2 The Modules

Now that we have discussed the motivation behind my work and the goals I want to achieve, we can discuss how I plan to achieve these goals.

To execute the content goals, I collaborated with Professor Janice Hudgings and Professor Dwight Whitaker of the physics department, who are teaching Physics 70 and Physics 71 in the 2024-25 academic year, respectively. Conversations with Prof. Hudgings began in mid-October, where she advised me to read through the *Six Ideas* textbook and tailor the mathematical concepts such that they tightly connected with what was presented in the

textbook. At this point, the students were about to begin learning about elementary quantum mechanics.

Through discussion of how to implement my work in their class, we decided that I would write a “module” discussing the mathematics behind concepts physics students were covering during a particular week of the semester. This would include a discussion of the mathematics, some enrichment content for interested students, and a short set of exercises to reinforce the material. The completed module would be incorporated into the students’ homework assignments for the week. Since the modules are oriented towards the mathematics behind physics, I found that using $\text{\LaTeX}$ to write the content was the most appropriate.

Ultimately, the modules were administered over the entirety of Unit Q in Physics 70 — one of three units taught that semester — to give a sense of completeness content-wise. More modules were initially considered for Unit T, the thermodynamics unit immediately succeeding Unit Q, but scrapped because only the first couple of sections of that unit were taught. Each week, I would send a first draft of my module to Prof. Hudgings about three days before the homework containing my modules were assigned. She would respond with feedback to implement about a day later, and I would finally send a polished draft to be published on the course website. We decided that responses would be submitted through Gradescope as the students’ “final question”.

In order to make the content as accessible as possible, I aimed for a multimodal learning method. While most of the content was written like a traditional textbook, I also included video animations of certain concepts to help students visualize the material and further benefit students who identify as visual learners (like myself). To gather relevant content, I combed through Professor Thomas Moore’s *Six Ideas* textbook in parallel with content that was covered week to week according to syllabi handed directly from Professors Hudgings and Whitaker [13]. I then made a list of the mathematical topics that were covered for each section, then picked one or two concepts I thought would most benefit from animation.

To create the animations, I used Python, in particular a library by Grant Sanderson (better known online as 3Blue1Brown) known as **Manim**. I was new to this library, so I combed through the documentation to gain a basic understanding of the workings as I programmed

my animations. Once the animation was rendered to my liking, I uploaded the video to YouTube and embedded it into the module. The code for the animations, the $\text{\LaTeX}$ for the module, and the videos can be found in the link pasted at the footnote [1](#). Figure [3.2](#) shows two stills from an animation I wrote about standing waves using the `Manim` library.

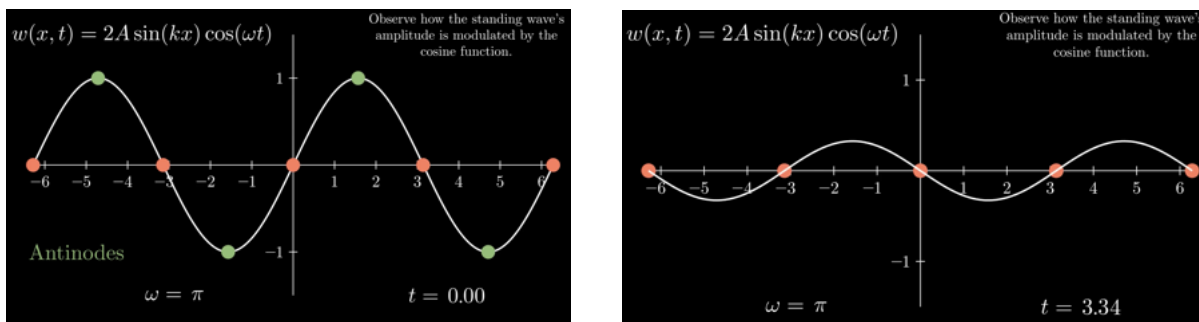


Figure 3.2: Two stills from an animation I wrote about standing waves in `Manim`.

All in all, each module was a self-contained document, usually about 4-5 pages long. It would contain the mathematical content distilled from a particular week’s physics content, animations in blaring red boxes for visual learners, and exercises to reinforce the material. However, to not overwhelm students, I only required students to complete one exercise from each module.

To qualitatively assess the efficacy of my modules, I conducted a survey within the module. At the top every module, a green box is displayed that asks that following:

“After reading the module and doing the associated activities, summarize, in one to two sentences, the most important concept you learned through the module. Please also state which graduating class you are in. If you have already seen everything in this module before, “nothing” is also a valid answer. Additionally, if anything was particularly confusing, please make a note of it here too.”

By asking students to summarize the most important concept they believe they learned, I achieve multiple things: the concept they summarize is reinforced, summarization helps learning by forcing students to paraphrase content, and I can gauge whether the learning goals I set through the modules were achieved. Additionally, by asking students to state

¹<https://github.com/chrisw011407/math-thesis>

their graduating class, I can filter out responses from students who are not first-years, since the modules are especially aimed at first-year physicists. However, in later analysis, I opted to keep second-year responses since a non-negligible portion of physics majors begin their journey in their second year.

3.3 Data Collection: A Pipeline

The data collection process lasted three weeks and spanned three homework sets. This was chosen intentionally to exactly coincide when they learned Unit Q, the introductory quantum mechanics unit. Insulating my collection period to one unit would give a sense of completeness. I opted to exhaustively collect data from both sections of Physics 70 to ensure as much standardization in the responses as possible, amounting to approximately 50 responses per module and totaling to about 150 responses.

The responses came in through Gradescope as a “solution” to their last homework problem, which focused on my modules. Most solutions were handwritten, so I digitized them into an Excel spreadsheet along with their graduation year. The students’ names were not reported on the spreadsheet to preserve anonymity. To digitize written responses, I turned to ChatGPT-4o for transcription, which I found to be sufficiently competent by cross-checking with my own transcriptions for 20 responses and finding that all of them aligned perfectly.

I then wrote a script in Python which took the Excel spreadsheet and partitioned the responses by homework. The code can be found in my Github and also in Appendix B. Post-partition, the code used the Natural Language Processing Toolkit (NLTK) package in Python to filter out stopwords — words in the English language that typically don’t serve much meaning in data analysis. The filtered responses were then concatenated and analyzed to see which words most frequently appeared. Two main deliverables came from this process: firstly, a qualitative and visual showcase of the responses by means of a word cloud, and secondly, a more quantitative analysis of how often words I listed in a word bank appeared in the responses. Each word in the word bank was chosen in accordance with the principles I wanted students to absorb from reading the module, so a comparison with the word bank would give me rudimentary intuition for how well students were digesting the content.

Chapter 4

Analysis

In the following subsections, I will break down each module, first guiding the reader through key features of the module and my rationale behind them. Subsequently, I will reveal the word bank (and thus implicitly the aims), followed by the word cloud and a quantitative table displaying the concept/phrase and number of occurrences, as well as the average number of matches per response. Modules for reference can be found in Appendix A.

At the beginning of each module, a brief description and instructions for the practice problems for the module are given: *“Note: Practice questions are denoted by a blue box. There are three practice questions in this module; you are only required to do one. Animations are denoted by a red box. You are strongly encouraged to watch them (they are also only about 40 seconds long each)! Sections that are labeled with “enrichment” are completely optional. Additionally, please answer the survey question immediately below.”* This is followed by a survey question contained in a green box, and I extensively used student answers to the survey question to qualitatively understand how well I conveyed my conceptual aims to the students.

With all subsequent modules, the assumption is that students have at least learned precalculus and are co-registered for Calculus I.

4.1 Module 1 – Waves and Boundary Conditions

In class, students were learning about fundamental wave mechanics and how to describe quantum mechanics using them. This module complemented that by elaborating on wave properties, with a table at the bottom of the first page that lists the different wave quantities as well as a short, intuitive description. The table was supplemented with a `manim` animation (hereon referred to as manimation) outlined in a red box, which visualized a few properties of waves as outlined in the table. Finally, the section on wave properties is wrapped up by a practice question that asks a two-part question. First, it quizzes the relationship of wavenumbers, angular frequency, and phase velocity, which aims to consolidate students' understanding of wave variables. It then asks about examining the relationship between momentum and (de Broglie) wavelength, re-establishing the physical connection of the mathematics. The goal of this section was to review students' knowledge of trigonometric functions and their properties.

The next section discussed superposition, standing waves, and boundary conditions. I explained superposition with the context of the wave equation. However, since differential equations was not assumed as prior knowledge, a further exploration into the wave equation was relegated to an enrichment section for students that did have more mathematical background and wanted to learn something new. I then provided another animation that illustrated what standing waves looked like, including annotations pointing out nodes and antinodes. Finally, I related these concepts to boundary conditions, introducing vocabulary such as Dirichlet Boundary Conditions and fundamental modes. I did not expect students to completely understand the significance of all vocabulary thrown at them, but having heard the words and a first-pass definition would ensure that they would have an easier time digesting subsequent encounters of the same vocabulary. The section ended with another practice question aimed to test students' understanding of boundary conditions, a relatively new concept that is not frequently discussed in pre-college mathematics but is integral to most physics problems.

The module has one final section on Taylor series, but it is completely enrichment material. Originally, it was included in the mandatory reading for the module, but I decided

to convert it to an enrichment section after further discussions with Prof. Hudgings. Since students were not expected to have knowledge beyond Calculus I, and Taylor series is typically covered in the later lectures of Calculus II, I figured it would be best to make it an optional section. For interested students, I included an example to illustrate the power of Taylor series and how they might have already seen it at play in earlier lectures, including the binomial approximation. Finally, a practice problem involving expanding Taylor series helps students understand the necessity of using these tools in physics and gives students a chance to practice expanding series.

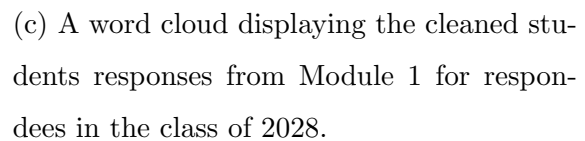
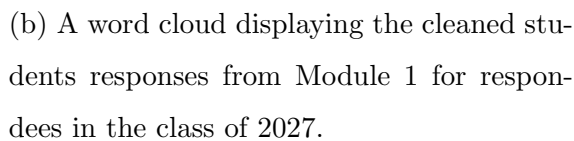
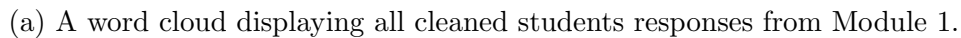
With a synopsis of the module, the word bank below highlights key concepts I hoped students would take away from the module. It is accompanied by a table that highlights the five most commonly matched terms from the word bank, which helps me understand which concepts students found most important.

Word Bank for Module 1

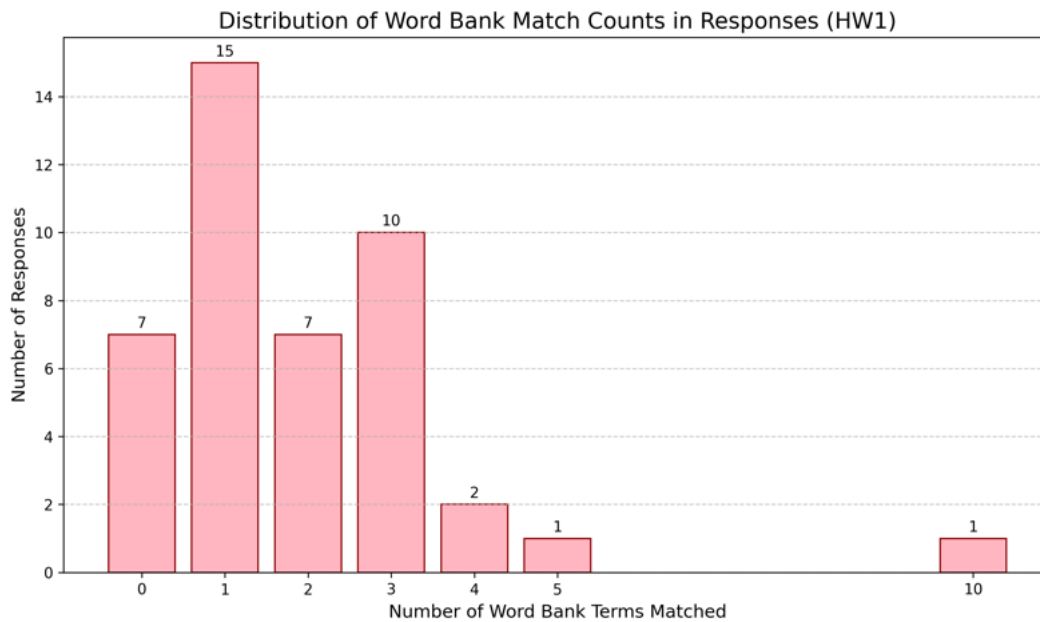
“traveling wave”, “wave equation”, “boundary condition”, “destructive interference”, “wave superposition”, “standing wave”, “fundamental mode”, “normal mode”, “taylor series”, “amplitude”, “wavelength”, “frequency”, “superposition”, “node”, “antinode”, “mode”, “harmonics”, “wave”, “waves”, “linear”, “k”, “omega”

Table 4.1: The five most commonly matched words or phrases from the word bank for Module 1.

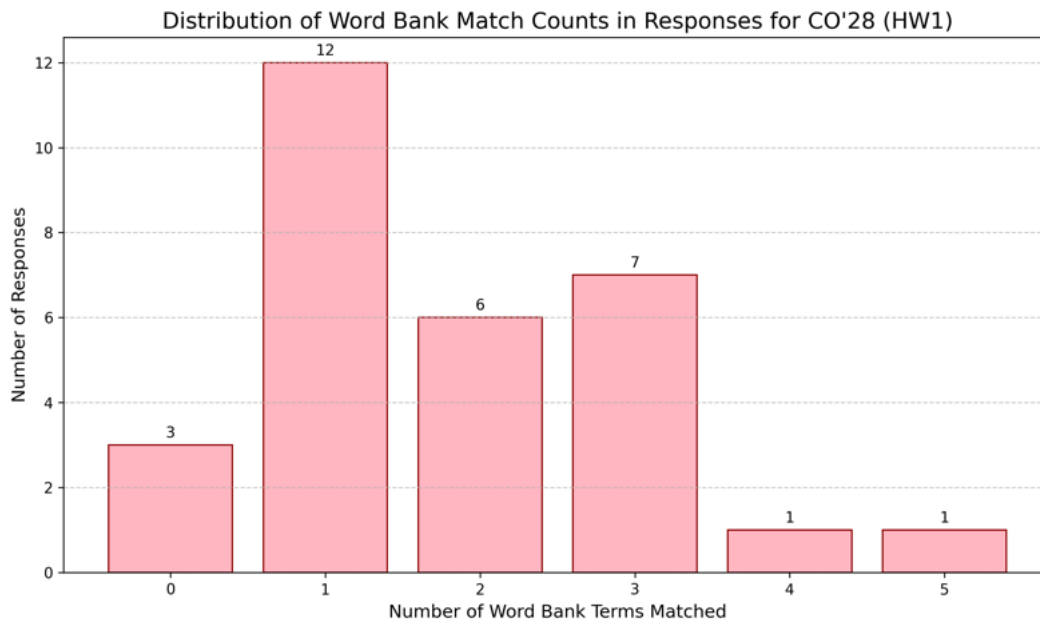
Phrase	Occurrences
boundary condition	11
wave	9
wave equation	8
taylor series	7
frequency	7



16



(a) A total match count plot of student responses for Module 1. The average number of matches was 1.91 matches per response.



(b) A match count plot of student responses for the class of 2028 only. The average number of matches per response was 1.80.

Figure 4.2: Match count plots of student responses for Module 1, visualizing how many word bank terms were matched to a student response.

Based on table [4.1](#) and figure [4.1](#), one can see that boundary conditions and waves (presumably wave properties) were most commonly cited as the most important concept taken away from the module, followed by the wave equation, Taylor series, and finally frequency — one of the wave properties. This was as I had hoped, since I thought the section which discussed novel mathematics most relevant to a budding physicist involved boundary conditions and how they were applied to solve problems.

Inspecting figure [4.2](#) for a more quantitative view of how helpful my modules were, it is apparent that most students took away concepts that appeared on my word bank, signaling an alignment of intention and delivery. Additionally, if we ignore the anomalous data point of 10 word bank terms matched, the first-years, on average, benefited from the modules more than the sophomores, with an average of 1.80 word bank terms matched per response in comparison to the 1.71 word bank terms matched per response for both the class of 2027 and 2028.

4.2 Module 2 – A Whiff of Linear Algebra

This module was designed to support students’ learning of linear algebra concepts fundamental to quantum mechanics. The module begins with “A Whiff of Linear Algebra,” establishing the connection between the newly encountered Dirac bra-ket notation and more familiar vector operations. By explicitly showing that inner products of quantum state vectors (when entries are real) are equivalent to dot products, I created a conceptual bridge between students’ existing knowledge of vector mathematics and the quantum mechanical formalism they were encountering for the first time. Real numbers are used explicitly because I do not assume familiarity with the complex numbers.

The mathematical relationship between inner products and dot products is emphasized deliberately to help students recognize that quantum mechanical formalism, though initially appearing abstract, builds upon mathematical structures they already understand. By drawing parallels between the dot product’s geometric interpretation (as a measure of “similarity” between vectors) and the inner product in quantum mechanics, I helped students develop an intuition for the physical meaning of quantum mechanical operations. This approach draws on the assumption that students have at least seen the dot product before, presumably in precalculus courses when vectors were first introduced.

The first practice question introduces students to a “mysterious physical system” with three states, requiring them to perform basic inner product calculations. This exercise serves multiple purposes: it reinforces computational fluency with inner products, it introduces the notion of state similarity through normalized inner products, and it prompts students to think about the mathematical bounds on these similarity measures. These concepts become crucial in more advanced quantum mechanics courses where students analyze probability amplitudes and transition probabilities between energy states.

The second practice problem directly connects the mathematical formalism to experimental data through the Stern-Gerlach apparatus, asking students to determine the angle θ from observed counts of spin states. This problem specifically aims to solidify the connection between the abstract vector space formalism and observable physical phenomena — a critical connection for physics education. By working backward from empirical results to mathe-

mathematical descriptions, students develop a deeper appreciation for how quantum theory makes testable predictions.

The enrichment section on complex numbers was included to support students who are curious about learning complex numbers more extensively than in their prior coursework. Since quantum mechanics relies heavily on complex amplitudes, this section prepares interested students for the more advanced material in subsequent lectures. The explanation of the Argand diagram provides a geometric intuition for complex numbers, which is particularly valuable for physics students who often think visually about abstract concepts.

The final practice problem challenges students to determine if two complex-valued quantum state vectors are orthogonal, reinforcing the Hermitian inner product concept and preparing students for upcoming material on eigenspaces and quantum measurement. This naturally extends to more advanced topics in upper-division quantum mechanics courses where students will encounter concept of orthogonal eigenstates more formally.

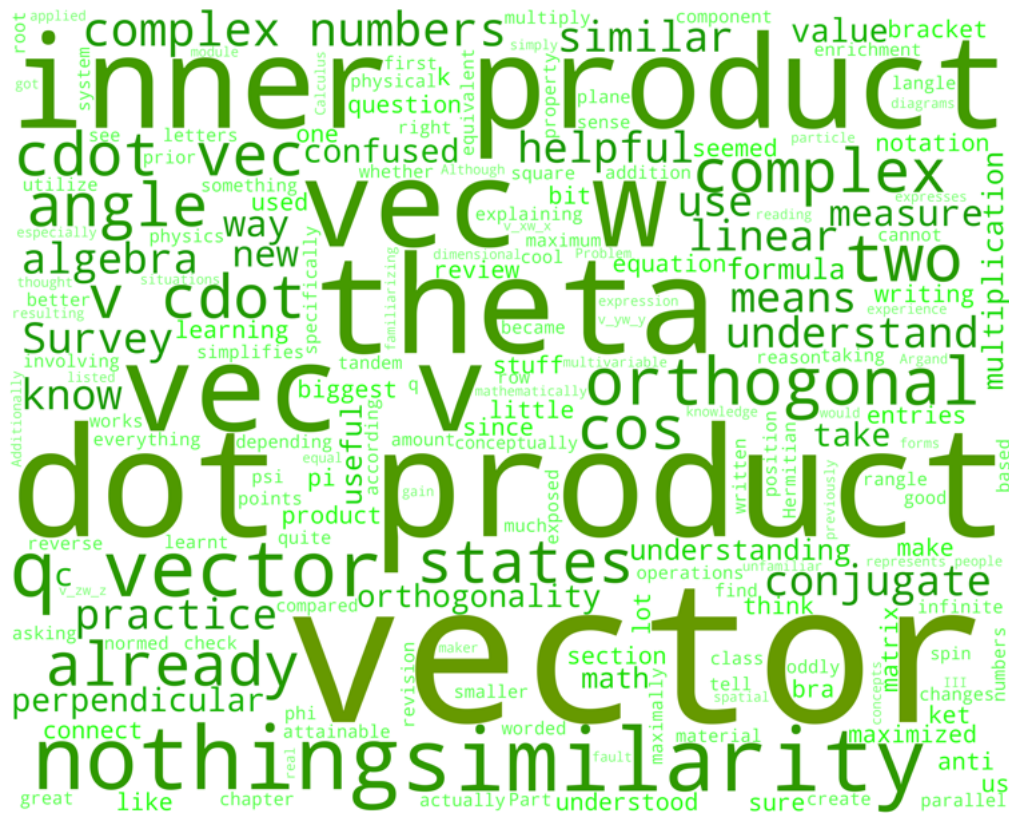
With a summary of what is in the module, we can now inspect the word bank for module 2 and its corresponding table, much like we did for the first module.

Word Bank for Homework 2

“inner product”, “dot product”, “Hermitian inner product”, “q-vector”, “vector space”, “Pythagorean theorem”, “complex conjugate”, “probability amplitude”, “complex number”, “imaginary number”, “q-vector”, “orthogonal”, “orthogonality”, “eigenvalue”, “eigenvector”, “wavefunction”, “probability”, “magnitude”, “amplitude”, “similarity”, “orthogonal”, “parallel”, “normalization”, “complex”, “conjugate”, “dimension”, “operator”, “vec”, “transpose”, “vector”, “vectors”, “similar”, “argand”

Table 4.2: The five most commonly matched words or phrases from the word bank for Module 2.

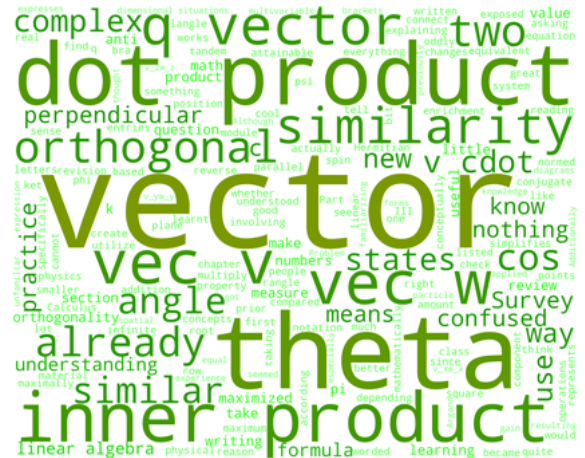
Phrase	Occurrences
vec	18
orthogonal	12
dot product	11
inner product	10
similarity	7



(a) A word cloud displaying all cleaned students responses from Module 2.

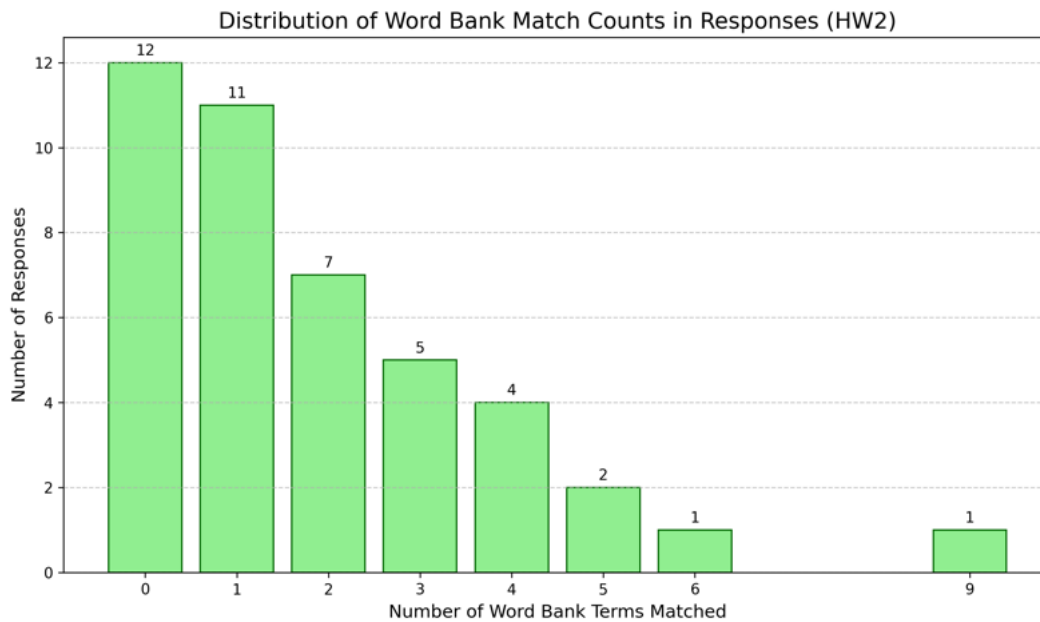


(b) A word cloud displaying the cleaned students responses from Module 2 for respondents in the class of 2027.

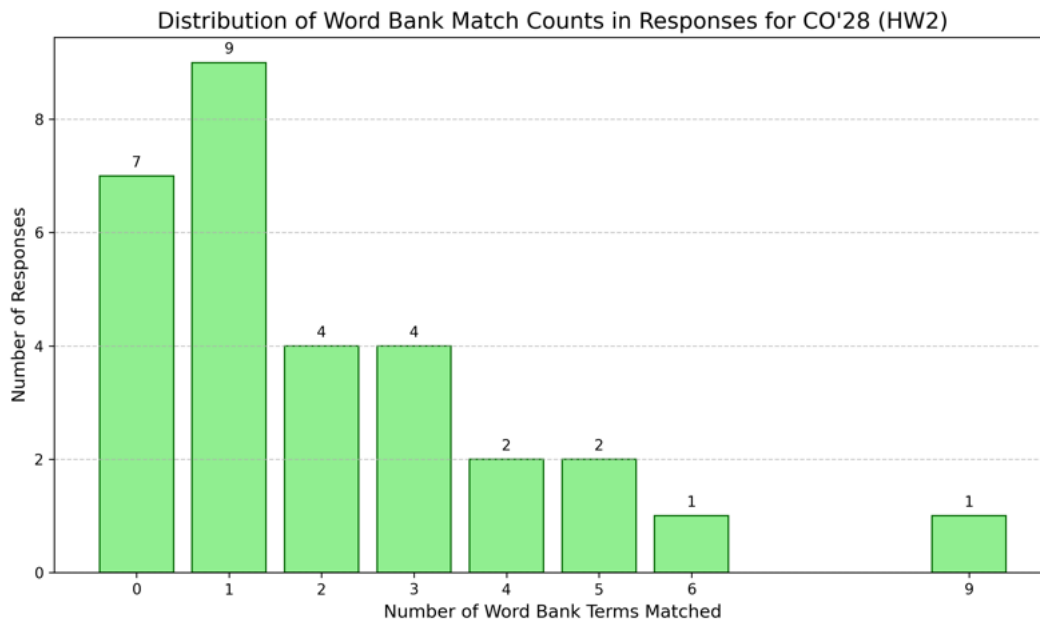


(c) A word cloud displaying the cleaned students responses from Module 2 for respondents in the class of 2028.

Figure 4.3: Word clouds for Module 2. Larger words correspond to more occurrences in student responses.



(a) A total match count plot of student responses for Module 2. The average number of matches was 1.70 matches per response.



(b) A match count plot of student responses for the class of 2028 only. The average number of matches per response was 2.07.

Figure 4.4: Match count plots of student responses for Module 2, visualizing how many word bank terms were matched to a student response.

Examining the data, I see that aside from mention of vectors, dot products and inner products are the most commonly cited important concepts, as well as a smaller subset of students that mentioned similarity. As these were mathematical concepts heavily emphasized in the physics the students were learning, I was pleased to find that many students picked up the importance of inner products, particularly their interpretation as a measure of similarity between two vectors. I also find it interesting that the word cloud shows complex numbers being cited more frequently in the sophomore respondees compared to the freshmen, as evidenced by figure [4.3](#). This can be explained by the fact that complex numbers are typically acquired into a student's mathematical arsenal later in their career.

A comparison of the subfigures in figure [4.4](#) reinforces the trend noticed in the first module; on average, students are taking something away from the module that aligns with topics I had intended to emphasize, and physicists in the graduating class of 2028 received more utility from the module, with 0.37 more word bank matches per response. Even excluding the anomalous data point with nine word bank terms matched, the first-year class average comes out to be 1.83 word bank terms matched per response, an increase of 0.13 over the overall result.

4.3 Module 3 – Probabilities and more Eigenstuff

The main purpose of this module is to supplement and extend students’ understanding of quantum mechanical wavefunctions, especially calculating and interpreting the overlap between two wavefunctions, colloquially called the “overlap integral”.

The opening section on normalizing wavefunctions provides a crucial bridge between the discrete quantum state vector formalism students recently learned and the continuous wavefunction representation. By deliberately showing how summation in the discrete case becomes integration in the continuous case, I help students with precalculus backgrounds make the conceptual leap to calculus-based physics. The explanation of probability amplitude as $\psi(x) = \langle x|\psi \rangle$ connects back to the inner product notation they learned previously, reinforcing that connection while introducing the new concept.

I included a brief enrichment section on Hilbert spaces to acknowledge the mathematical depth behind these concepts without requiring students to master it at this stage. This approach allows mathematically curious students to glimpse the more advanced framework while preventing overwhelm for those focusing on building foundational understanding. Students who continue to upper-division quantum mechanics will encounter Hilbert spaces more formally, and this introduction plants a seed for future learning.

The first practice problem challenges students to normalize a specific wavefunction using a definite integral. This problem was carefully chosen because it requires only the power rule for integration, which should be accessible to students in calculus I. The setup encourages students to apply their developing integration skills in a physics context, reinforcing both the mathematical technique and its physical interpretation. The second part of the problem, asking about the limit as n approaches infinity, invites students to think about the asymptotic behavior of functions — a concept that becomes increasingly important in advanced physics courses, for instance when students are asked to find a potential function at infinity.

The section on boundary conditions introduces students to eigen-concepts (eigenvalues, eigenvectors, and eigenfunctions), which are fundamental to quantum mechanics but often challenging for students encountering them for the first time. Recognizing that students may not

have seen these concepts in their linear algebra coursework yet, I provided a self-contained explanation that relies only on their precalculus background. The included manimation visualizes eigenvectors, leveraging students' visual intuition to complement the mathematical formalism.

The second practice problem extends boundary conditions to a new scenario, asking students to reason about which energy levels would be prohibited if a potential barrier were placed at the center of the box. This problem reinforces the connection between nodes in standing waves, mathematical concepts introduced in the first module, and the quantization of energy levels. The hint connecting back to the cello example provides a tangible reference point.

The final section on calculating probabilities brings the module full circle, applying the inner product concept from the previous module to continuous wavefunctions. By explicitly showing how $|\langle\phi|\psi\rangle|^2$ gets expressed as an integral in position space, I help students understand how the abstract quantum formalism connects to concrete calculations they perform throughout their quantum mechanics courses. Following that, an enrichment section on function space decomposition introduces the powerful concept of Fourier decomposition — a topic students will encounter more formally in their upper-division coursework, most extensively in junior-level electromagnetism.

The final practice problem reinforces integration techniques while developing physical intuition about wavefunction overlap. By suggesting that students graph the functions, I encourage them to connect the mathematical results with visual representations—an approach that utilizes their precalculus background in function plotting. As with the previous two modules, we can now contextualize the word bank and subsequent table.

Word Bank for Homework 3

“wavefunction”, “normalization condition”, “Hilbert space”, “vector space”, “infinite-dimensional space”, “probability amplitude”, “eigenfunction”, “energy eigenvalue”, “energy eigenstate”, “energy level”, “Hamiltonian operator”, “quantum system”, “boundary condition”, “potential well”, “infinite potential”, “overlap integral”, “particle in a box”, “Fourier decomposition”, “completeness relation”, “wavefunction”, “eigenfunction”, “eigenvalue”, “eigenvalues”, “eigenvector”, “eigenvectors”, “normalization”, “normalized”, “position”, “momentum”, “continuum”, “operator”, “orthogonal”, “Hamiltonian”, “potential”, “eigenstates”, “boundary”, “integral”, “probability”, “probabilities”, “transformation”, “transformations”

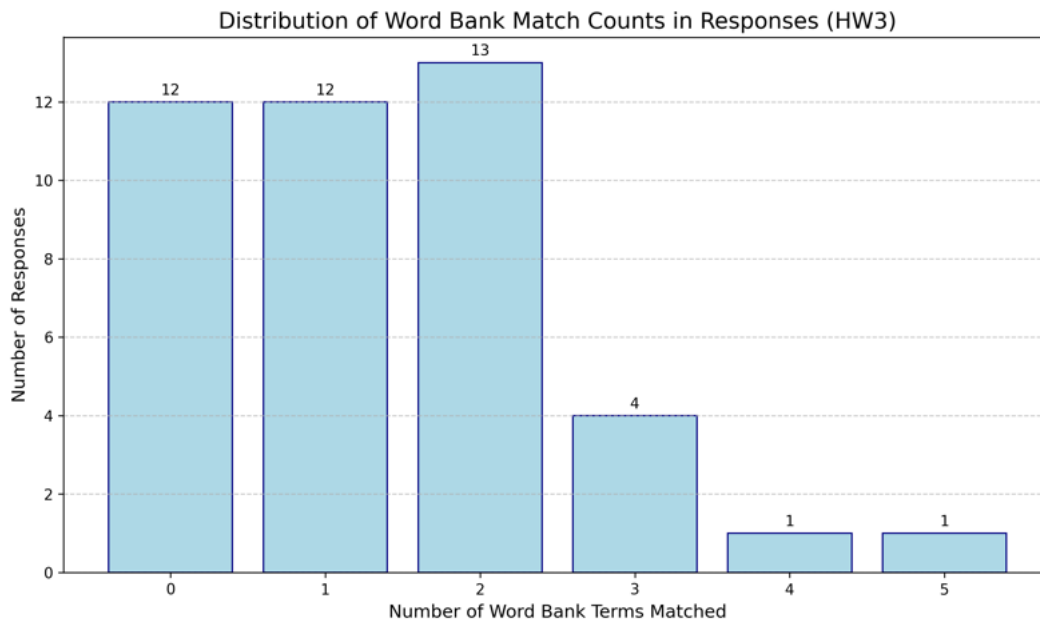
Table 4.3: The five most commonly matched words or phrases from the word bank for Module 3.

Phrase	Occurrences
probability	8
hilbert space	5
eigenvalues	4
hamiltonian operator	4
eigenvectors	4

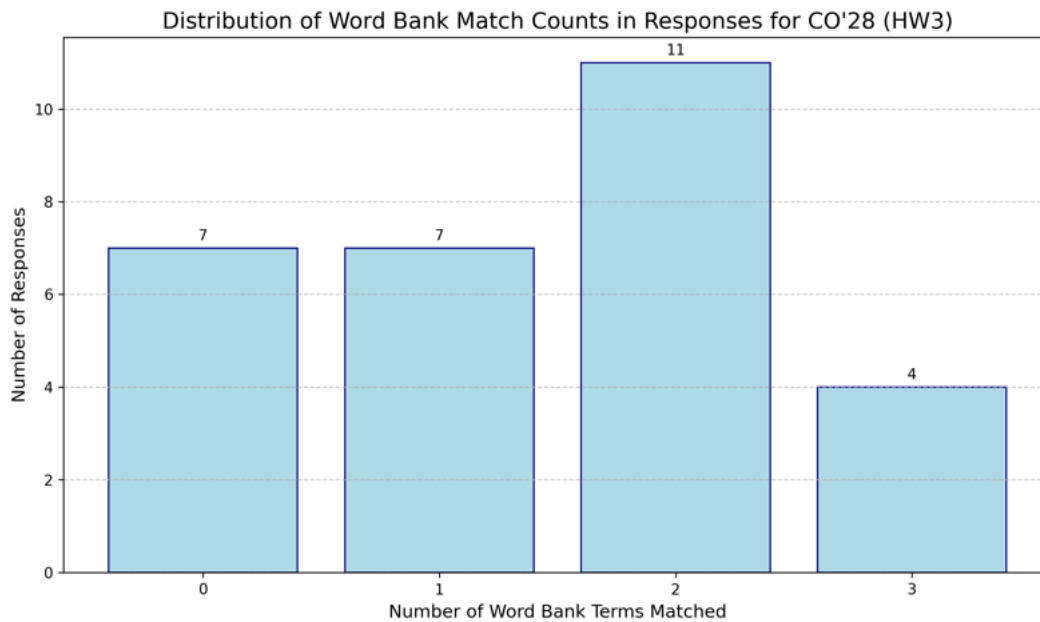
Table 4.3 and figure 4.5 align mostly with expectations, with probabilities and eigen-concepts being crucial and foundational concepts covered in the module to extend students’ linear algebraic backgrounds and connect it to the continuous case, where integration is involved. Particularly surprising was the appearance of “Hilbert Space” as the *second* most common term — intended purely as an enrichment module, it seemed as though more people read and determined that information to be significant than expected.

The match count plots on figure 4.6 reaffirm the observed trend in the prior two modules; the combination of sophomores and freshmen averaged 1.37 matches to the word bank per response, with a marginal improvement upon exclusively evaluating freshmen up to 1.41 matches per response.

The observed consistent difference in benefit received from the modules between sophomores



(a) A total match count plot of student responses for Module 3. The average number of matches was 1.37 matches per response.



(b) A match count plot of student responses for the class of 2028 only. The average number of matches per response was 1.41.

Figure 4.6: Match count plots of student responses for Module 1, visualizing how many word bank terms were matched to a student response.

and freshmen can be explained by the simple fact that sophomores have had more time in college. Consequently, they have had more time to expose themselves to college-level mathematics and thus familiarize themselves with the mathematics used in Physics 70. Therefore, they will likely receive less benefit from a math module.

Chapter 5

Recommendations and Future Work

The mathematical modules developed in this thesis have shown promise in supporting first-year physics students, particularly those in the class of 2028. Based on the analysis in Chapter 4, students consistently identified key mathematical concepts from the modules that aligned with my pedagogical intentions. Moving forward, there are several recommendations for how this work could be expanded and improved.

5.1 Integration into the Physics Curriculum

The current implementation of these modules has been tightly integrated with Physics 70's quantum mechanics unit. However, to create lasting impact within the physics department, a more systematic approach to incorporation is needed. I recommend embedding relevant mathematical modules throughout the introductory physics sequence (Physics 70, 71, 72), with exercises becoming regular components of weekly problem sets.

Since the professors teaching the introductory sequence may rotate year-to-year, developing a comprehensive yet versatile resource would be more sustainable than modules tailored to specific instructional styles. I propose developing an expanded document that functions as a mathematical companion text for first-year physics courses. This document would be organized chronologically to align with when students encounter various mathematical tools in their physics coursework.

For example, a chapter on Taylor series could be positioned early in the document, coinciding

with Unit R (special relativity) when students first encounter the binomial approximation. Later chapters could introduce multivariable calculus concepts like derivative operators when students reach Physics 72’s electromagnetism content. Each chapter would maintain the successful elements of the current modules: practice questions, `manim` animations for visual learners, and optional enrichment sections.

This approach would give instructors flexibility to assign specific chapters or sections as supplementary reading based on their pedagogical priorities and the particular needs of their students. The requirement to complete one practice problem from the assigned section would ensure student engagement while not overwhelming their workload.

5.2 Enhanced Data Collection and Analysis

While the current analysis provides encouraging qualitative evidence for the efficacy of these mathematical supplements, future iterations should incorporate more robust data collection methods to better understand the relationship between students’ mathematical backgrounds and the benefits they derive from the modules.

The survey question asking students to identify the most important concept they learned allowed for a basic assessment of alignment between teaching goals and student takeaways. However, I have no concrete evidence to verify my hypothesis that students who answered “nothing” on the survey had stronger mathematical preparation prior to college. Future iterations should include additional survey questions to gather data on students’ mathematical backgrounds, potentially including:

1. Their highest level of mathematics completed before college.
2. Whether they had prior exposure to calculus, linear algebra, or differential equations.
3. Their self-assessed comfort level with mathematical concepts in physics.
4. Specific areas of mathematics they find challenging in the physics context.

With this additional data, correlations between prior mathematical preparation and benefit from the modules could be analyzed more systematically. This analysis would inform further

refinement of the resource, particularly in determining the appropriate depth and breadth of the enrichment sections.

The mathematical modules developed in this thesis represent a promising first step toward more comprehensive support for first-year physics students. With careful integration into the curriculum and enhanced data collection, this approach has the potential to strengthen the mathematical foundations of physics students at Pomona College and contribute to a more inclusive physics education, ensuring that mathematics does not pose a barrier to those that want to learn physics.

Appendix A

Modules & Solutions

The following pages contain all of the math modules I have written, along with solutions to the practice problems.

A Module on the Maths behind Q1,Q2,Q3,Q5

Chris Wang

October 29, 2024

Note: Practice questions are denoted by a blue box. There are three practice questions in this module; you are only required to do one. Animations are denoted by a red box. You are strongly encouraged to watch them (they are also only about 40 seconds long each)! Sections that are labeled with “enrichment” are completely optional. Additionally, please answer the survey question immediately below.

SURVEY QUESTION

After reading the module and doing the associated activities, summarize, in one to two sentences, the most important concept you learned through the module. If you have already seen everything in this module before, “nothing” is also a valid answer.

Traveling Waves

There are a number of important quantities associated with traveling wave, whose “wiggle function” is presented again below:

$$w(t, x) = A \sin(kx - \omega t) \quad (\text{linear wave})$$

Below, we have a table of important quantities associated with the wave:

Quantity	Intuition
A	Amplitude of the wave, how “tall” the wave is.
f	Frequency of the wave, or how many cycles of a wave passes through a point per second.
ω	Angular frequency, $\omega = 2\pi f$. Like frequency, but measures the angular “distance” a wave elapses per second in radians / second.
λ	Wavelength of the wave, or the distance between two crests of the wave.
k	Wave number, $k = 2\pi/\lambda$. Essentially, how many wave cycles you can fit into some (unit) length.
$ \vec{v} $	Wave speed or phase velocity, $ \vec{v} = f\lambda = \omega/k$ gives an idea of how fast the wave appears to travel.
T	Period of the wave, $T = \frac{1}{f}$, gives a sense of how long it takes for a wave to complete a cycle through a certain point.

ACTIVITY

Watch the animation → [here](#), which visualizes a few properties of waves seen above.

PRACTICE 1

- (a) Intuitively explain why, if we double the wavenumber (k) of a wave while keeping the angular frequency (ω) constant, the phase velocity ($|\vec{v}|$) halves. *Think about the definition of angular velocity and how the wavenumber is visually related to wavelength.*
- (b) Momentum of a particle is typically given by $p = h/\lambda = \hbar k$, where k is the wave number and $\hbar = h/2\pi$. If the momentum of a particle is doubled, what do you expect happens to its (de Broglie) wavelength?

Superposition, Standing Waves, and Boundary Conditions

Most waves a physicist deals with in the same context will be superposable, meaning we can meaningfully add two traveling waves together to get a new wave. The mathematical reason behind this is that the algebraic form of these waves are solutions of something called the **wave equation**, which is a second-order partial differential equation that describes how waves propagate.

THE WAVE EQUATION (ENRICHMENT)

While this is not something you need to worry about for now, if you are interested, the form of the wave equation is

$$\frac{\partial^2 w}{\partial t^2} = v^2 \frac{\partial^2 w}{\partial x^2},$$

where w describes the wave solution and v is the wave speed. If you are familiar with differential equations and want to try an extra problem, show that the wave function $w(t, x) = A \sin(kx - \omega t) + B \sin(kx + \omega t)$ is a solution to the wave equation.

It turns out, as you will learn in a differential equations course, that solutions to the wave equation are linear, meaning that if $w_1(t, x)$ and $w_2(t, x)$ are solutions to the wave equation, then $c_1 w_1(t, x) + c_2 w_2(t, x)$ is also a solution to the wave equation (c_1, c_2 are real numbers). This is why we can add waves together and get a new wave. The principle that describes this is known as the **superposition principle**.

Under special circumstances, two traveling waves can add to form a **standing wave**. Recall from Q2 that such a standing wave can be formed by an incident wave and reflected wave that are superposed. In particular,

$$w_{\text{incident}} = A \sin(kx - \omega t) \quad (\text{incident wave})$$

$$w_{\text{reflected}} = A \sin(kx + \omega t), \quad (\text{reflected wave})$$

which gives us the standing wave

$$w_{\text{standing}} = 2A \sin(kx) \cos(\omega t). \quad (\text{standing wave})$$

To help further visualize this, complete the activity below.

ACTIVITY

Some properties of the standing wave, in particular its oscillatory nature as determined by $\cos(\omega t)$, is shown in this → [animation](#).

Before we go further, it is useful to review some vocabulary: for any standing wave, the **nodes** are the points where the wave is always zero, while the **antinodes** are the points where the wave oscillates between its extrema.

Fixing the nodes and antinodes of a standing wave can lead to important results. For example, if a string is fixed to both ends, like a guitar or cello string, the ends of the string must be nodes: the string is restricted from moving around at the ends. This is an example of a **boundary condition**, which is a condition that must be satisfied by the wave at the boundaries of the system. To introduce the vocabulary, the aforementioned boundary condition falls into a category of boundary conditions known as **Dirichlet boundary conditions**. More generally, Dirichlet boundary conditions are conditions that specify the value of the wave at the boundaries of the system.

Because nodes ($\sin(n\pi) = 0$ for an integer n) and antinodes ($\sin(n\pi + \pi/2) = \pm 1$) of the wave correspond to multiple solutions, we can have multiple standing waves that satisfy the boundary conditions. The set of all possible standing waves that satisfy the boundary conditions is known as the **normal modes** of the system, where the solution corresponding to the lowest frequency is known as the **fundamental mode**, and all higher modes, which are integer multiples of the fundamental frequency, are known as the **harmonics**.

Let's navigate an example: consider a pipe organ like the one mentioned in the *Six Ideas* textbook, particularly a pipe of length L . If both ends of the pipe are either closed or open, then the ends of the pipe must both be nodes (if closed) or antinodes (if open). This means that this restriction implies that the length of the pipe corresponds to half a wavelength, or $\lambda/2$. This means that the fundamental frequency of the pipe is $f = v/2L$, where v is the speed of sound in the pipe. We can also satisfy these conditions if we multiply the frequency by a natural number, resulting in harmonics that are integer multiples of the fundamental frequency.

PRACTICE 2

The A string on a cello produces a fundamental frequency of 220 Hz. If I gently put my finger somewhere on the string, I can generate an ethereal 880 Hz sound. If the length of the string is L , where on the string am I putting my finger? (*Hint: think about the boundary conditions of the string and which of the harmonics satisfy these boundary conditions. There are two possible answers.*)

A Taster of Taylor Series (Enrichment)

Taylor series are an extremely important of physics, as they allow us to approximate functions using polynomials, which are generally much easier to work with. In general, the Taylor series of a function $f(x)$ about a point a is given by

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n, \text{ (Taylor Series)}$$

where $f^{(n)}(a)$ denotes the n th derivative of f evaluated at a , and $n! = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot 2 \cdot 1$ denotes n factorial. The Taylor series is a polynomial approximation of $f(x)$ about a , and the more terms we include, the better the approximation. Note that most of the utility of Taylor series comes from truncating the series at some point, which is allowed if $x - a \ll 1$.

There is an intuitive way to understand the Taylor series. Remember that, in approximating a function with a polynomial, we want to ensure that the function's behavior is identical to its polynomial approximation. This means that the approximation should have the same value as the function at the point of interest (zeroth derivative), the same slope (first derivative), the same curvature (second derivative), and so on. If you would like to, convince yourself the formula above does satisfy this criterion by taking and matching successive derivatives of the Taylor series with the derivatives of the function at $x = a$.

You may have already seen Taylor series at work secretly. For example, the binomial approximation

$$(1+x)^n \simeq 1+nx$$

is a Taylor series approximation of $(1+x)^n$ about $x=0$ truncated at the linear term. Its full expansion is

$$(1+x)^n = 1+nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots = \sum_{k=0}^{\infty} \frac{n!}{k!(n-k)!}x^k.$$

To really show off the binomial approximation, let's calculate 1.001^{100} and compare it to the binomial approximation. First, we have

$$1.001^{100} \simeq 1.105$$

as the true value. The binomial approximation gives us

$$1+100(0.001) = 1.1,$$

giving us an error of about 0.45%. Not bad for a linear approximation!

PRACTICE 3 (ENRICHMENT)

The reason I brought up Taylor series here is because the Rayleigh criterion mentioned in Q3 relates the diffraction limit of an aperture to the wavelength of light and the aperture size *nonlinearly*. The nonlinear nature can make calculations annoying, especially at smaller angular separations. The Rayleigh criterion is given by

$$\sin \theta = 1.22 \frac{\lambda}{D},$$

where θ is the angular separation between two objects of interest, λ is the wavelength of light, and D is the diameter of the aperture.

- Taylor expand $\sin \theta$ about $\theta = 0$ to first order and find the minimum aperture diameter D that can resolve two objects separated by $\theta = 1$ arcsecond using light of wavelength $\lambda = 500$ nm.
- For larger values of θ , truncating the series at first order may be insufficient. Find the next nonzero term in the Taylor series for $\sin \theta$ and compare the error between the series in (a) containing only one term, the new series, and the true value for $\theta = \pi/4$.

Solutions for HW9

Chris Wang

October 30, 2024

PRACTICE 1

- (a) Intuitively explain why, if we double the wavenumber (k) of a wave while keeping the angular frequency (ω) constant, the phase velocity ($|\vec{v}|$) halves. *Think about the definition of angular velocity and how the wavenumber is visually related to wavelength.*
- (b) Momentum of a particle is typically given by $p = h/\lambda = \hbar k$, where k is the wave number and $\hbar = h/2\pi$. If the momentum of a particle is doubled, what do you expect happens to its (de Broglie) wavelength?

Solution:

- (a) Recall that the phase velocity of a wave is

$$|\vec{v}| = \frac{\omega}{k}.$$

Hence, the velocity is inversely proportional to the wavenumber and proportional linearly to the angular frequency. If we double the wavenumber, we see (based on the animation) that the wavelength halves, so the distance between two crests is also halved. Now, if the angular frequency remains constant, the same time elapses between two crests passing the same point. Hence, if only half the distance is traversed in the same time interval, the phase velocity must be halved.

- (b) The equation $p = h/\lambda$ is most useful here. Since h is a constant, doubling the momentum only changes the (de Broglie) wavelength. In this case, the wavelength is halved.

PRACTICE 2

The A string on a cello produces a fundamental frequency of 220 Hz. If I gently put my finger somewhere on the string, I can generate an ethereal 880 Hz sound. If the length of the string is L , where on the string am I putting my finger? (*Hint: think about the boundary conditions of the string and which of the harmonics satisfy these boundary conditions. There are two possible answers.*)

Solution:

The cello string is a case where both ends are fixed; hence, they are nodes. In that case, 220 Hz corresponds to half a cycle (the fundamental frequency). By putting my finger on the string, I essentially induce another node somewhere along the string that forces the new “fundamental frequency” to be 880 Hz. In essence, the node provides a boundary condition that the new wave must satisfy. We can see a figure that illustrates this below:



Figure 1: On the left: The fundamental frequency of the string. On the right: The new “fundamental frequency” once my finger is placed on the string.

We also observe that the new frequency is four times the original fundamental frequency, which implies that the new wavelength is a quarter of the original wavelength. Hence, the finger is placed at $L/4$ or $3L/4$.

PRACTICE 3

The reason I brought up Taylor series here is because the Rayleigh criterion mentioned in Q3 relates the diffraction limit of an aperture to the wavelength of light and the aperture size *nonlinearly*. The nonlinear nature can make calculations annoying, especially at smaller angular separations. The Rayleigh criterion is given by

$$\sin \theta = 1.22 \frac{\lambda}{D},$$

where θ is the angular separation between two objects of interest, λ is the wavelength of light, and D is the diameter of the aperture.

- Taylor expand $\sin \theta$ about $\theta = 0$ to first order and find the minimum aperture diameter D that can resolve two objects separated by $\theta = 1$ arcsecond using light of wavelength $\lambda = 500$ nm.
- For larger values of θ , truncating the series at first order may be insufficient. Find the next nonzero term in the Taylor series for $\sin \theta$ and compare the error between the series in (a) containing only one term, the new series, and the true value for $\theta = \pi/4$.

Solution:

- To first order around $\theta = 0$, the Taylor expansion of $\sin \theta$ is

$$\sin \theta \simeq \sin(0) + \left. \frac{d \sin \theta}{d\theta} \right|_{\theta=0} \theta = 0 + \cos(0) \theta = \theta.$$

We can now find D by algebraically isolating the equation above:

$$\theta = 1.22 \frac{\lambda}{D} \quad \Rightarrow \quad D = 1.22 \frac{\lambda}{\theta}$$

Note that 1 arcsecond is $1/(3600 * 57.3)$ radians. Hence, $D = 1.22 \times 5 \times 10^{-7} / (1/(3600 * 57.3)) \approx 0.13$ m.

- Let's repeat the above expansion, bringing it further:

$$\sin \theta \simeq \theta + \underbrace{\frac{d^2 \sin \theta}{d\theta^2} \bigg|_{\theta=0} \frac{\theta^2}{2!}}_0 + \frac{d^3 \sin \theta}{d\theta^3} \bigg|_{\theta=0} \frac{\theta^3}{3!} = \theta - \frac{\theta^3}{3!}.$$

Let's compare for $\theta = \pi/4$:

$$\sin(\pi/4) = \sqrt{2}/2 \approx 0.7071, \quad \theta = \pi/4 \approx 0.7854, \quad \theta - \frac{\theta^3}{3!} \approx 0.7046.$$

Evidently, adding in next term in the expansion greatly improves its accuracy, even for relatively larger angular displacements away from $\theta = 0$.

THE WAVE EQUATION (ENRICHMENT)

While this is not something you need to worry about for now, if you are interested, the form of the wave equation is

$$\frac{\partial^2 w}{\partial t^2} = v^2 \frac{\partial^2 w}{\partial x^2},$$

where w describes the wave solution and v is the wave speed. If you are familiar with differential equations and want to try an extra problem, show that the wave function $w(t, x) = A \sin(kx - \omega t) + B \sin(kx + \omega t)$ is a solution to the wave equation.

Solution:

Plug our trial solution into the wave equation:

$$\frac{\partial^2 w}{\partial t^2} = -A\omega^2 \sin(kx - \omega t) - B\omega^2 \sin(kx + \omega t)$$

$$\frac{\partial^2 w}{\partial x^2} = -Ak^2 \sin(kx - \omega t) - Bk^2 \sin(kx + \omega t)$$

Observe that since $v^2 = \omega^2/k^2$:

$$\begin{aligned} v^2 \frac{\partial^2 w}{\partial x^2} &= -\frac{\omega^2}{k^2} (Ak^2 \sin(kx - \omega t) + Bk^2 \sin(kx + \omega t)) \\ &= -A\omega^2 \sin(kx - \omega t) - B\omega^2 \sin(kx + \omega t) \\ &= \frac{\partial^2 w}{\partial t^2}. \end{aligned}$$

A Module on the Maths behind Q6-7

Chris Wang

November 7, 2024

Note: Practice questions are denoted by a blue box. There are three practice questions in this module; you are only required to do one. Animations are denoted by a red box. Sections that are labeled with “enrichment” are completely optional. Additionally, please answer the survey question immediately below.

SURVEY QUESTION

After reading the module and doing the associated activities, summarize, in one to two sentences, the most important concept you learned through the module. If you have already seen everything in this module before, “nothing” is also a valid answer. Additionally, if anything was particularly confusing, please make a note of it here too.

A Whiff of Linear Algebra

In Q7, you encountered the beginnings of Dirac’s bra-ket notation for quantum mechanics. To briefly review, recall the notion of a **q-vector** for **bras** and **kets**:

$$\text{A ket is } |\psi\rangle = \begin{bmatrix} \psi_1 \\ \psi_2 \\ \vdots \end{bmatrix}, \quad \text{while a bra is } \langle\psi| = \begin{bmatrix} \psi_1^* & \psi_2^* & \cdots \end{bmatrix}.$$

The funny stars at the top right of the bra vector denote the complex conjugate of the entries in the ket vector. However, for the purposes of this module, we will focus on when these entries are purely real numbers, or when $a^* = a$.

We will see that, with this restriction, the **inner product** of these two vectors is equivalent to the dot product between the two vectors. Recall that, for two vectors $\vec{v}$ and $\vec{w}$, the dot product is given by:

$$\vec{v} \cdot \vec{w} = v_x w_x + v_y w_y + v_z w_z.$$

This is the same as the inner product of two q-vectors, $\langle\psi|$ and $|\phi\rangle$, which is given by:

$$\langle\psi|\phi\rangle = \begin{bmatrix} \psi_1 & \psi_2 & \cdots \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \end{bmatrix} = \psi_1 \phi_1 + \psi_2 \phi_2 + \psi_3 \phi_3 + \cdots = \sum_i \psi_i \phi_i.$$

The difference here is that the q-vectors can have any number of entries depending on the physical system being studied, while the spatial vectors $\vec{v}$ and $\vec{w}$ are strictly three-dimensional (since we live in three-dimensional space). For instance, if we were looking at q-vectors corresponding to the electron’s spin in

the z -direction, we would have a two-dimensional q -vector, since the electron can either be spin up or spin down.

Essentially, for two q -vectors, each with k components, their inner product is equivalent to the dot product in k -dimensional space!

Now that we have established their equivalence, let's see how the dot product or inner product can be interpreted as a measure of "similarity" between two vectors. It turns out that the dot product can be written in another way:

$$\vec{v} \cdot \vec{w} = |\vec{v}| |\vec{w}| \cos(\theta),$$

where θ is the angle between the two vectors and $|\cdot|$ represents the magnitude of a vector, or its length. If the two vectors are parallel, then $\theta = 0$, and the dot product is maximized since $\cos \theta = 1$. If the two vectors are perpendicular, then $\theta = \pi/2$, and the dot product is zero. In this case, the vectors are known to be **orthogonal**. If $\theta = \pi$, the dot product is *minimized*, implying the vectors have maximum "anti-similarity". This is the same for the inner product of two q -vectors: if the two vectors are parallel or "maximally similar", the inner product is maximized, while if the two vectors are orthogonal, the inner product is zero.

What happens if you take the inner product of a q -vector with itself? Well, let's appeal to its equivalence to the dot product. If we take the dot product of a vector with itself, we get:

$$\vec{v} \cdot \vec{v} = v_x^2 + v_y^2 + v_z^2 = |\vec{v}|^2.$$

You may recognize this as the three-dimensional analogue of the Pythagorean theorem. Indeed, the square length of a vector is the sum of the squares of its components. This is the same for the inner product of a q -vector with itself, where we may think of the equation below as a k -dimensional analogue of the Pythagorean Theorem:

$$\langle \psi | \psi \rangle = \sum_i \psi_i^2 = |\psi|^2.$$

To enforce the equivalence between the dot product and the q -inner product, let's look at a few exercises.

PRACTICE 1

Consider a mysterious physical system that can take on three physical states: A, B, C . State $|\alpha\rangle$ has two particles in state A and one particle in state B , so

$$|\alpha\rangle = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

Meanwhile, state $|\beta\rangle$ has one particle in state B and two particles in state C , so

$$|\beta\rangle = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

PRACTICE 1

- Calculate $\langle \alpha | \alpha \rangle$. Remember that $\langle \alpha |$ is the row vector version of $|\alpha\rangle$.
- Calculate $\langle \beta | \beta \rangle$.
- How similar are the two states $|\alpha\rangle$ and $|\beta\rangle$? Essentially, what is

$$\frac{\langle \alpha | \beta \rangle}{\sqrt{\langle \alpha | \alpha \rangle \langle \beta | \beta \rangle}}? \quad (1)$$

- What is the maximum attainable value of the expression in Eq. 1? Consider arbitrary q-vectors $|\alpha\rangle$ and $|\beta\rangle$. *Hint: you can make qualitative or quantitative arguments here.*

PRACTICE 2

A beam of electrons are sent through a Stern-Gerlach apparatus, like the SG machines in Q7. The beam is initially in some state $|+\theta\rangle$ as in Table Q7.1. As it passes through an SG_z detector, the following data is recorded:

State	Counts
Spin up	75
Spin down	25

- What is θ ? *Hint: the probabilities are **normalized**, meaning $|\langle +\theta | +z \rangle|^2 + |\langle +\theta | -z \rangle|^2 = 1$.*
- What would the data look like if the initial state was instead $|-x\rangle$?

Everything below this sentence is enrichment material.

AN INTRODUCTION TO COMPLEX NUMBERS (ENRICHMENT)

Complex numbers were first invented by an Italian mathematician named Gerolamo Cardano in the 16th century to help find a general solution to cubic equations. The key ingredient to complex numbers is the imaginary number i , which by definition satisfies

$$i^2 = -1.$$

Having this, we can now define a complex number z as

$$z = a + bi,$$

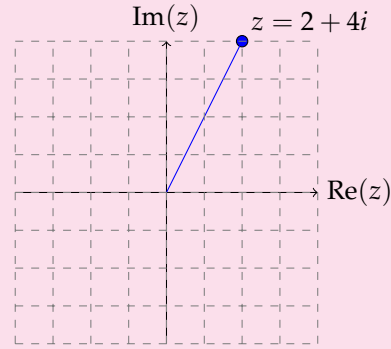
where a and b are real numbers. The real part of z is denoted by $\text{Re}(z)$, while the imaginary part of z is denoted by $\text{Im}(z)$. The complex conjugate of z , denoted by z^* , is given by

$$z^* = a - bi.$$

A special property of complex numbers and their conjugate is that the product of a complex number and its conjugate is always real:

$$zz^* = (a + bi)(a - bi) = a^2 + b^2 = |z|^2,$$

where $|z|$ is the magnitude of the complex number z . One can intuitively imagine the complex plane as a two-dimensional plane, where the x -axis is the real part of the complex number and the y -axis is the imaginary part. Thus, we can “plot” any complex number on this plane. Such a diagram is called an **Argand diagram**. To illustrate, we can plot the complex number $z = 2 + 4i$ on the Argand diagram, as shown on the right:



Now that we have established the nature of bras and kets, we can now introduce the **inner product** of two vectors. In regular Euclidean space, the inner product of two vectors is the dot product, as highlighted above. However, with complex numbers, the inner product is slightly different, and is given the name **Hermitian inner product**. The Hermitian inner product of two vectors $\langle\psi|$ and $|\phi\rangle$ is given by

$$\text{For } \langle\psi| = [\psi_1^* \quad \psi_2^* \quad \cdots] \text{ and } |\phi\rangle = \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \end{bmatrix}, \quad \langle\psi|\phi\rangle = \sum_i \psi_i^* \phi_i.$$

You may notice that if we reverse the order of the vectors, we have

$$\langle\phi|\psi\rangle = \sum_i \phi_i^* \psi_i = \left(\sum_i \psi_i^* \phi_i \right)^* = \langle\psi|\phi\rangle^*.$$

This is a property unique to the Hermitian inner product: the regular (Euclidean) inner product does not have this property, since all real numbers satisfy $a = a^*$.

AN INTRODUCTION TO COMPLEX NUMBERS (ENRICHMENT)

As outlined in Q7.3, there are a couple more properties and vocabulary to take note of. If we take the inner product of a q-vector with itself, we get

$$\langle \psi | \psi \rangle = \sum_i \psi_i^* \psi_i = \sum_i |\psi_i|^2.$$

This looks a lot like the *square* of the magnitude of the q-vector. Indeed! It turns out that the inner product of a q-vector with itself is equal to the square of the q-vector's magnitude.

Also recall that the angle between two vectors can be found using the dot product for the Euclidean case. Using this as a heuristic, we see that if $\theta = \pi/2$, the dot product must be zero, since $\cos(\pi/2) = 0$. This translates to the vectors being perpendicular, or more generally, **orthogonal** to each other. In the context of quantum mechanics, we say that two vectors are orthogonal if their (Hermitian) inner product is zero.

PRACTICE 3

Consider the two q-vectors $|\psi\rangle = \begin{bmatrix} 1+i \\ -1 \end{bmatrix}$ and $|\phi\rangle = \begin{bmatrix} 1+i \\ 2 \end{bmatrix}$. Are they orthogonal?

Solutions for HW10

Chris Wang

November 7, 2024

PRACTICE 1

Consider a mysterious physical system that can take on three physical states: A, B, C . State $|\alpha\rangle$ has two particles in state A and one particle in state B , so

$$|\alpha\rangle = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

Meanwhile, state $|\beta\rangle$ has one particle in state B and two particles in state C , so

$$|\beta\rangle = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}.$$

- a) Calculate $\langle\alpha|\alpha\rangle$. Remember that $\langle\alpha|$ is the row vector version of $|\alpha\rangle$.
- b) Calculate $\langle\beta|\beta\rangle$.
- c) How similar are the two states $|\alpha\rangle$ and $|\beta\rangle$? Essentially, what is

$$\frac{\langle\alpha|\beta\rangle}{\sqrt{\langle\alpha|\alpha\rangle\langle\beta|\beta\rangle}}? \tag{1}$$

- d) What is the maximum attainable value of the expression in Eq. 1? Consider arbitrary q-vectors $|\alpha\rangle$ and $|\beta\rangle$. *Hint: you can make qualitative or quantitative arguments here.*

Solution:

a)

$$\langle \alpha | \alpha \rangle = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = 2^2 + 1^2 + 0^2 = |\alpha|^2 = 5.$$

b)

$$\langle \beta | \beta \rangle = \begin{bmatrix} 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = 0^2 + 1^2 + 2^2 = |\beta|^2 = 5.$$

c) The reason we have the denominator is to normalize the inner product. With this in mind, we have that

$$\frac{\langle \alpha | \beta \rangle}{\sqrt{\langle \alpha | \alpha \rangle \langle \beta | \beta \rangle}} = \frac{\langle \alpha | \beta \rangle}{|\alpha| |\beta|}$$

Now, we can calculate the numerator:

$$\langle \alpha | \beta \rangle = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} = 2(0) + 1(1) + 0(2) = 1.$$

Therefore,

$$\frac{\langle \alpha | \beta \rangle}{\sqrt{\langle \alpha | \alpha \rangle \langle \beta | \beta \rangle}} = \frac{1}{\sqrt{5}\sqrt{5}} = \frac{1}{5}.$$

d) The maximum value of the expression in Eq. 1 can be obtained when we just set the q-vectors to be the same (this does indeed reflect maximal similarity!). In this case, we have that

$$\frac{\langle \alpha | \alpha \rangle}{\sqrt{\langle \alpha | \alpha \rangle \langle \alpha | \alpha \rangle}} = \frac{|\alpha|^2}{\sqrt{|\alpha|^2 |\alpha|^2}} = \frac{|\alpha|^2}{|\alpha|^2} = 1.$$

PRACTICE 2

A beam of electrons are sent through a Stern-Gerlach apparatus, like the SG machines in Q7. The beam is initially in some state $|+\theta\rangle$ as in Table Q7.1. As it passes through an SG_z detector, the following data is recorded:

State	Counts
Spin up	75
Spin down	25

- a) What is θ ? *Hint: the probabilities are **normalized**, meaning $|\langle +\theta|+z\rangle|^2 + |\langle +\theta|-z\rangle|^2 = 1$.*
- b) What would the data look like if the initial state was instead $| -x \rangle$?

Solution:

- a) Using the hint, we first calculate

$$\langle +\theta|+z\rangle = \cos(\theta/2), \quad \langle +\theta|-z\rangle = \sin(\theta/2).$$

One can see that the normalization condition is a good sanity check, as

$$|\langle +\theta|+z\rangle|^2 + |\langle +\theta|-z\rangle|^2 = \cos^2(\theta/2) + \sin^2(\theta/2) = 1.$$

However, we also know that the probabilities are

$$P(+z) = |\langle +\theta|+z\rangle|^2 = \cos^2(\theta/2) = \frac{75}{100} = \frac{3}{4}, \quad P(-z) = |\langle +\theta|-z\rangle|^2 = \sin^2(\theta/2) = \frac{25}{100} = \frac{1}{4}.$$

With this in mind, we can solve for θ , which technically yields two solutions. We will focus on one for now, where

$$\theta = 2 \cos^{-1} \left(\sqrt{\frac{3}{4}} \right) = 2 \cos^{-1} \left(\frac{\sqrt{3}}{2} \right) = 2 \times \frac{\pi}{6} = \frac{\pi}{3}.$$

Another valid solution is $\theta = -\pi/3$.

- b) If the initial state was $| -x \rangle$, then the probabilities would be

$$P(+z) = |\langle -x|+z\rangle|^2 = \frac{1}{2}, \quad P(-z) = |\langle -x|-z\rangle|^2 = \frac{1}{2}.$$

Therefore, the data would be

State	Counts
Spin up	50
Spin down	50

PRACTICE 3

Consider the two q-vectors $|\psi\rangle = \begin{bmatrix} 1+i \\ -1 \end{bmatrix}$ and $|\phi\rangle = \begin{bmatrix} 1+i \\ 2 \end{bmatrix}$. Are they orthogonal?

Solution:

Let's take their (Hermitian) inner product:

$$\langle\psi|\phi\rangle = \begin{bmatrix} 1-i & -1 \end{bmatrix} \begin{bmatrix} 1+i \\ 2 \end{bmatrix} = (1-i)(1+i) + (-1)(2) = 1+i-i+1-2 = 0.$$

Since the inner product is zero, the two q-vectors are orthogonal.

A Module on the Maths behind Q9 and QB

Chris Wang

November 12, 2024

Note: Practice questions are denoted by a blue box. There are three practice questions in this module; you are only required to do one. Animations are denoted by a red box. Sections that are labeled with “enrichment” are completely optional. Additionally, please answer the survey question immediately below.

SURVEY QUESTION

After reading the module and doing the associated activities, summarize, in one to two sentences, the most important concept you learned through the module. Please also state which graduating class you are in (this is to help with some survey results). If you have already seen everything in this module before, “nothing” is also a valid answer. Additionally, if anything was particularly confusing, please make a note of it here too.

Normalizing Wavefunctions

In Q9, you saw how we could express a q-vector in continuous parameter spaces (wavefunctions) and some properties associated with the wavefunctions, such as the collapse of the wavefunction, normalization, and the Heisenberg Uncertainty Principle.

First, let’s quickly recap going from q-vectors to wavefunctions. In general, q-vectors describe the possible states of a quantum system, like how we had two-component q-vectors that described spin states. We can also describe q-vectors for position and momentum space, except they would be infinite-dimensional, as seen in the *Six Ideas* text; they are infinite-dimensional because we can have an infinite number of possible positions or momenta, these spaces are continuous. In this continuum picture, the q-vectors are more conveniently described by wavefunctions, and the inner product $\langle x|\psi\rangle$ can be more easily interpreted as $\psi(x)$, the so-called “probability amplitude” of finding our state $|\psi\rangle$ at position x .

HILBERT SPACES (ENRICHMENT)

The q-vectors live in a type of mathematical space called a vector space (if you’ve taken or are taking linear algebra, you’re familiar with this). However, familiar vector spaces are all finite, and strange things, especially regarding keeping inner products finite, happen when we venture into the infinite. To make things mathematically rigorous, mathematicians formalized a special type of space called a **Hilbert Space**, which does address convergence issues that arise in infinite-dimensional spaces. The position and momentum spaces that you encounter in Q9 are examples of Hilbert Spaces.

Let's now segue into normalization. Remember the dot product between two q-vectors and how its magnitude squared can be interpreted as a probability? If we add up probabilities of a q-vector being in all states of a physical system, we should get 1. After all, the state q-vector has to take on some value! This is the idea behind normalization.

Let's take the z-spin q-vector as an example. If I have an arbitrary q-vector $|\psi\rangle$ in this physical system, I know it has to either be spin up or down. Therefore $P(\psi \rightarrow +z) + P(\psi \rightarrow -z) = 1$, where $P(\psi \rightarrow +z) = |\langle +z|\psi\rangle|^2$ and $P(\psi \rightarrow -z) = |\langle -z|\psi\rangle|^2$ (review Q7.4 rule 4 if you are not convinced). In this formalism,

$$|\langle +z|\psi\rangle|^2 + |\langle -z|\psi\rangle|^2 = 1.$$

Extending this to our position space, we *integrate* instead of sum over the states (remember that integration is the continuum analogue of discrete summation), yielding the corresponding normalization condition for our wavefunction:

$$\int_{-\infty}^{\infty} |\langle x|\psi\rangle|^2 dx = \int_{-\infty}^{\infty} \underbrace{\psi(x)\psi^*(x)}_{|\psi(x)|^2} dx = 1.$$

Thankfully, the wavefunctions we're interested in are all real, meaning $|\psi(x)|^2 = \psi(x)^2$.

PRACTICE 1

Suppose a wavefunction in position space is given by

$$\psi(x) = \begin{cases} A(1 - (x/L)^{2n}) & \text{if } |x| \leq L \\ 0 & \text{otherwise.} \end{cases}$$

- Supposing $\psi(x)$ is normalized and n is a positive integer, find the value of A . *Hint: the power law for integrals states that $\int \alpha x^n dx = \alpha x^{n+1}/(n+1)$. Let $\alpha = L^{-2n}$ and integrate from $-L$ to L .*
- What value does A approach as $n \rightarrow \infty$? You can deduce this by taking the limit of A , which from (a) should depend on n .

More Boundary Conditions

A Brief Note on Eigen-stuff

Before we transition to discussing the specific math behind QB, let's briefly discuss eigenvalues, eigenvectors, and eigenfunctions. This box is especially helpful if you haven't worked with these things in a linear algebraic context. The prefix "eigen" comes from the German word for "own" or "characteristic", and we can think of determining eigenvalues, eigenvectors, and eigenfunctions as finding values, vectors, or functions that are characteristic of some transformation.

What does that mean? For **eigenvectors**, we seek vectors that don't change direction when we apply a transformation or operation on them. Essentially, the post-operation vector is just a scaled version of the pre-operation vector. The scaling factor is known as the **eigenvalue**. For **eigenfunctions**, we're looking for functions that don't change form when we apply an operator to them. The eigenvalue is the value that the operator returns when applied to the eigenfunction.

For instance,

$$\text{if } \vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \text{ and } M\vec{v} = \begin{bmatrix} -2 \\ -4 \end{bmatrix} = -2\vec{v} \text{ for some matrix } M,$$

then we say $\vec{v}$ is an eigenvector of M with eigenvalue -2 . Similarly, if $f(x) = \sin(kx)$, and the Δ operator corresponded to taking the derivative with respect to x twice, then $\Delta f(x) = -k^2 f(x)$, so $\sin(kx)$ is an eigenfunction of the Δ operator with eigenvalue $-k^2$.

Applying this to our particular context, the functions that you work with in QB have the form

$$\psi_{E_n}(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \text{ when } 0 \leq x \leq L \text{ and } 0 \text{ otherwise.}$$

These are eigenfunctions of the **Hamiltonian operator** $\hat{H}$ with **energy eigenvalue** E_n , meaning the equation $\hat{H}\psi_{E_n}(x) = E_n\psi_{E_n}$ holds. The Hamiltonian operator $\hat{H}$ is the operator version of the Hamiltonian H , which is a quantity that tells us the total energy (kinetic + potential) of the system. Hence, it makes sense why the operator's eigenvalues are E_n ! The Hamiltonian is something you will explore in more detail in upper-division mechanics.

MORE ON THE HAMILTONIAN OPERATOR (ENRICHMENT)

For those of you who have more experience with differential equations, you can explicitly show that the equation above holds, which is known as the (time-independent) **Schrödinger equation**. The Hamiltonian operator for the box has the form

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \text{ when } 0 \leq x \leq L,$$

and using the fact that $\frac{d^2}{dx^2} \sin(kx) = -k^2 \sin(kx)$, you can show that $\hat{H}\psi_{E_n}(x) = E_n\psi_{E_n}$, where E_n is as in QB.6b.

For a visual on how eigenvectors behave, complete the activity below.

ACTIVITY

Watch the animation → [here](#), which illustrates how eigenvectors (and non-eigenvectors) behave under a transformation. Note that eigenvalues can also be negative, in which case the transformation “flips” the eigenvector.

Now that we have the basic form of our eigenfunctions, we can discuss the boundary conditions they must satisfy. In fact, Prof. Moore derives the coefficient within the sine functions by imposing the boundary conditions on the wavefunction (equations QB.3 and QB.4). For our quanton in a box, or any other quantum system with an infinite potential well outside a finite spatial range, our wavefunction must be zero outside the box and at its boundary. Intuitively, if the potential energy required to be in such regions is infinite, there's no way the quanton can be there!

PRACTICE 2

Suppose we have a quanton in a box, just like in QB. If we hypothetically imagined the potential to spike to infinity exactly halfway between the walls of the box (and remain zero elsewhere), which of the energy levels would now be *prohibited*? Why? *Hint: refer to Figure QB.3 for some intuition. Also, this is slightly similar to the cello example from two weeks ago.*

Calculating The Probability of Finding a Quanton in an Eigenstate

In Q7.4, Prof. Moore discusses the rules of quantum mechanics. Rule 4, in particular, tells us that the probability of finding a quanton $|\psi\rangle$ in an observable eigenstate $|a_n\rangle$ is $|\langle a_n|\psi\rangle|^2$. We can extend this: if we prepare two quantons $|\psi\rangle$ and $|\phi\rangle$, the probability of finding $|\psi\rangle$ in the state $|\phi\rangle$ is $|\langle\phi|\psi\rangle|^2$. For a set of observable eigenstates $|a_k\rangle$, where k ranges from 1 to n , such a probability can be rewritten:

$$|\langle\psi|\phi\rangle|^2 = \left| \sum_{k=1}^n \langle\psi|a_k\rangle \langle a_k|\phi\rangle \right|^2$$

Intuitively, this gives us a sense of how “similar” $|\psi\rangle$ and $|\phi\rangle$ are. After all, $|\langle\psi|\phi\rangle|^2$ is the magnitude squared of the dot product of the two vectors.

We close off this module with an extension to the continuum: finding the probability of two wavefunctions “overlapping”. This is sometimes called the “overlap integral”, which intuitively tells us how similar two wavefunctions are. In the context of QB, we can use this to find the probability of finding a quanton in a particular energy eigenstate. Using q-vector notation, the probability of finding a *normalized* quanton $|\psi\rangle$ in another *normalized* state $|\phi\rangle$ is nothing more than the magnitude squared of the inner product $\langle\phi|\psi\rangle$, or $|\langle\phi|\psi\rangle|^2$.

If we write this in position space, we have

$$|\langle\phi|\psi\rangle|^2 = \left| \int_{-\infty}^{\infty} \langle\phi|x\rangle \langle x|\psi\rangle dx \right|^2 = \left| \int_{-\infty}^{\infty} \phi^*(x)\psi(x) dx \right|^2.$$

Of course, in relevant cases, our wavefunctions are real, so $\phi^*(x) \mapsto \phi(x)$.

FUNCTION SPACE DECOMPOSITION (ENRICHMENT)

It turns out that we can write any arbitrary function defined over a finite interval (either $[0, L]$ or $[-L, L]$) as a sum of the energy eigenfunctions of a quanton in a box. This sum may be infinite, but it’s a powerful result that allows us to decompose any function into its constituent energy eigenstates. This is known as a **Fourier Decomposition**. This is possible by the **completeness relations** of the energy eigenfunctions. Generally, the completeness relations tell us that a set of basis vectors (or, in this case, functions) can span the entire relevant vector/function space – our function space is the space of periodic functions in this case. The completeness relations for position space also allowed us to write the inner product as we did right above Practice 1, and you will see more of the details when you take an upper level course in quantum mechanics.

PRACTICE 3

In QB.D2, you're asked to prove that the normalization constant for all quanton in a box energy eigenstates are $\sqrt{2/L}$. For this question, let's assume $L = 1$ and assume the prior statement is true.

(a) Let

$$\phi(x) = \begin{cases} \sqrt{30}x(1-x) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise,} \end{cases}$$

which is normalized (you don't have to prove this). Set up the integral that will allow you to find the probability of $\phi(x)$ being in the $n = 1$ energy eigenstate of our particle in a box.

(b) Find the probability of $\phi(x)$ being in the $n = 2$ energy eigenstate for our particle in a box. Feel free to use an integral calculator to evaluate the resulting integral. If it helps, graph the functions $\phi(x)$ and $\psi_{E_2}(x)$. Can you rationalize the result?

A Module on the Maths behind Q9-10

Chris Wang

November 12, 2024

PRACTICE 1

Suppose a wavefunction in position space is given by

$$\psi(x) = \begin{cases} A \sqrt{1 - (x/L)^{2n}} & \text{if } |x| \leq L \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Supposing $\psi(x)$ is normalized and n is a positive integer, find the value of A . *Hint: the power law for integrals states that $\int \alpha x^n dx = \alpha x^{n+1}/(n+1)$. Let $\alpha = L^{-2n}$ and integrate from $-L$ to L .*
- (b) What value does A approach as $n \rightarrow \infty$? You can deduce this by taking the limit of A , which from (a) should depend on n .

Solution:

- (a) We know that

$$\int_{-\infty}^{\infty} A^2 (1 - (x/L)^{2n}) dx = 1.$$

Evaluating the integral, we get

$$\begin{aligned} A^2 \int_{-L}^L (1 - (x/L)^{2n}) dx &= A^2 \left(x - \frac{x^{2n+1}}{L^{2n}(2n+1)} \right) \Big|_{-L}^L \\ &= A^2 \left(2L - \frac{L}{2n+1} + \frac{(-L)^{2n+1}}{L^{2n}(2n+1)} \right) \\ &= A^2 \left(2L - \frac{2L}{2n+1} \right) \\ &= 2A^2 L \left(\frac{2n}{2n+1} \right) = 1. \end{aligned}$$

Therefore,

$$A^2 = \frac{2n+1}{4nL} \implies A = \sqrt{\frac{2n+1}{4nL}}.$$

- (b) As $n \rightarrow \infty$, we see that $2n+1 \simeq 2n$, and so $A \simeq \sqrt{1/(2L)}$.

PRACTICE 2

Suppose we have a quanton in a box, just like in QB. If we hypothetically imagined the potential to spike to infinity exactly halfway between the walls of the box (and remain zero elsewhere), which of the energy levels would now be *prohibited*? Why? *Hint: refer to Figure QB.3 for some intuition. Also, this is slightly similar to the cello example from two weeks ago.*

Solution:

If the potential spikes to infinity halfway between the walls, we have another boundary condition to be aware of: the wavefunction must vanish at the midpoint. Inspecting Figure QB.3, we see that odd energy levels of the particle in a box **do not** have nodes at the midpoint, while even energy levels **do**. Therefore, the odd energy levels would be prohibited.

Mathematically, we can also prove this: since the $n = 2$ energy level is the first energy level to have a node at the midpoint, all integer multiples of this energy level will also have nodes at the midpoint, satisfying this new boundary condition. Therefore, the odd energy levels are prohibited.

PRACTICE 3

In QB.D2, you're asked to prove that the normalization constant for all quanton in a box energy eigenstates are $\sqrt{2/L}$. For this question, let's assume $L = 1$ and assume the prior statement is true.

(a) Let

$$\phi(x) = \begin{cases} \sqrt{30}x(1-x) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise,} \end{cases}$$

which is normalized (you don't have to prove this). Set up the integral that will allow you to find the probability of $\phi(x)$ being in the $n = 1$ energy eigenstate of our particle in a box.

(b) Find the probability of $\phi(x)$ being in the $n = 2$ energy eigenstate for our particle in a box. Feel free to use an integral calculator to evaluate the resulting integral. If it helps, graph the functions $\phi(x)$ and $\psi_{E_2}(x)$. Can you rationalize the result?

Solution:

(a) Our probability is $|\langle \phi | \psi_{E_1} \rangle|^2$, which can be written in integral form as

$$|\langle \phi | \psi_{E_1} \rangle|^2 = \left| \int_0^1 \langle \phi | x \rangle \langle x | \psi_{E_1} \rangle dx \right|^2 = \left| \int_0^1 \sqrt{30}x(1-x) \cdot \sqrt{2} \sin(\pi x) dx \right|^2.$$

(b) The probability of $\phi(x)$ being in the $n = 2$ energy eigenstate is

$$|\langle \phi | \psi_{E_2} \rangle|^2 = \left| \int_0^1 \sqrt{30}x(1-x) \cdot \sqrt{2} \sin(2\pi x) dx \right|^2.$$

Using an integral calculator, we find $|\langle \phi | \psi_{E_2} \rangle|^2 = 0$. We see that, graphically, over the interval $[0, 1]$, the function $\phi(x)$ looks like a parabola symmetric about $x = 1/2$, while ψ_{E_2} changes sign at $x = 1/2$ and appears to "cancel itself out". Multiplying these functions yields another function that appears to "cancel itself out" over the unit interval, which is why the probability is zero.

Appendix B

Appendix B – Computer Code for Cleaning and Processing Data

```
import pandas as pd
import os
import random
from PIL import Image
import numpy as np
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import RegexpTokenizer
from wordcloud import WordCloud
import matplotlib.pyplot as plt
# nltk.download('punkt')
# nltk.download('wordnet')
# nltk.download('omw-1.4')
# nltk.download('punkt_tab')
# from nltk.tokenize import word_tokenize

def red_color_func(word, font_size, position, orientation,
random_state=None, **kwargs):
    """
```

Generates a color function where:

- Largest words are deep, vibrant red (low lightness, high saturation).
- Smaller words are slightly more magenta (higher hue).
- Smaller words are lighter, and larger words are darker.

```
"""
```

```
hue = max(0, min(330, 330 - font_size * 0.05))
```

```
saturation = min(100, max(30, font_size * 2))
```

```
lightness = min(85, max(30, 100 - font_size // 2))
```

```
return "hsl(%d, %d%%, %d%%)" % (hue, saturation, lightness)
```

```
def green_color_func(word, font_size, position, orientation,  
random_state=None, **kwargs):
```

```
    hue = max(0, min(120, 120 - font_size * 0.05))
```

```
    saturation = min(100, max(30, font_size * 2))
```

```
    lightness = min(85, max(30, 100 - font_size // 2))
```

```
    return "hsl(%d, %d%%, %d%%)" % (hue, saturation, lightness)
```

```
def blue_color_func(word, font_size, position, orientation,  
random_state=None, **kwargs):
```

```
    hue = max(0, min(250, 250 - font_size * 0.05))
```

```
    saturation = min(100, max(30, font_size * 2))
```

```
    lightness = min(85, max(30, 100 - font_size // 2))
```

```

        return "hsl(%d, %d%%, %d%%)" % (hue, saturation, lightness)

# Load data
data = pd.read_csv('P70 responses.csv', encoding='latin1')
hw1 = data[['Graduating Year (1)', 'Response (1)']]
      .sort_values('Graduating Year (1)', ascending=False).dropna()

hw2 = data[['Graduating Year (2)', 'Response (2)']]
      .sort_values('Graduating Year (2)', ascending=False).dropna()

hw3 = data[['Graduating Year (3)', 'Response (3)']]
      .sort_values('Graduating Year (3)', ascending=False).dropna()

hw1 = hw1.drop(hw1[hw1['Graduating Year (1)'] <= 2026].index)
      .reset_index(drop=True)

hw2 = hw2.drop(hw2[hw2['Graduating Year (2)'] <= 2026].index)
      .reset_index(drop=True)

hw3 = hw3.drop(hw3[hw3['Graduating Year (3)'] <= 2026].index)
      .reset_index(drop=True)

stopwords = stopwords.words('english')
custom_word_list = ['learned', 'important', 'concept', 'really', 'thing', 'seen',
                    'liked', 'also', 'helps', 'made', 'helped', 'x', 'understand', 'helpful']
stopwords.extend(custom_word_list)
stop_words = set(stopwords) # Stop words

cloud_words = ''

```

```

for i in range(len(hw3)): # adjust for homework number
    response = hw3['Response (3)'][i]

    # Tokenize
    tokenizer = RegexpTokenizer(r'\w+')
    tokens = tokenizer.tokenize(response)

    filtered_sentence = [w for w in tokens if not w.lower() in stop_words]
    cloud_words += ' '.join(filtered_sentence) + ' '

# print(cloud_words)

wc = WordCloud(width = 4000, height = 3200,
               background_color = 'white',
               stopwords = stop_words,
               min_font_size = 10,
               color_func=blue_color_func).generate(cloud_words)

plt.figure(figsize = (10, 8), facecolor = None)
plt.imshow(wc)
plt.axis("off")
plt.tight_layout(pad = 0)
plt.savefig('wordcloud_hw3.png',dpi=500)
plt.show()

```


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